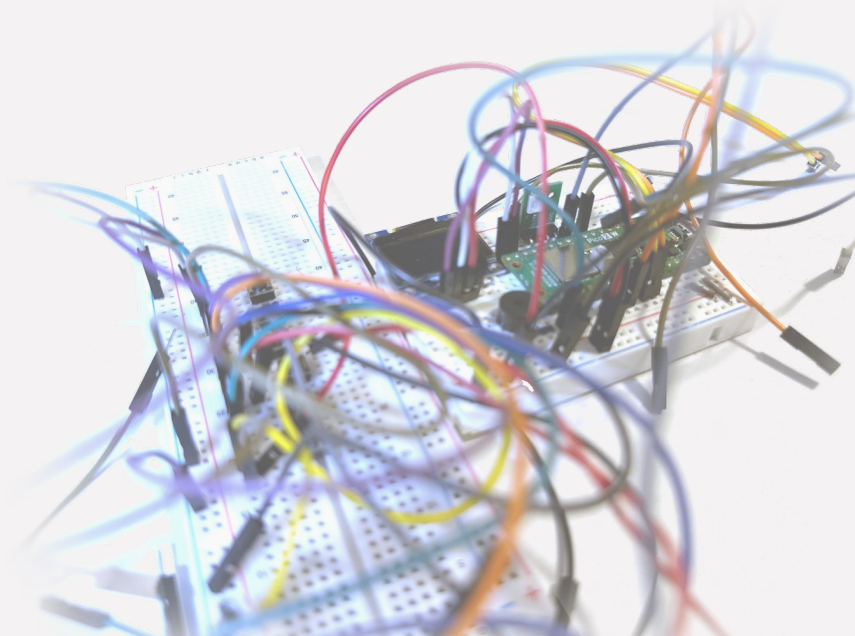


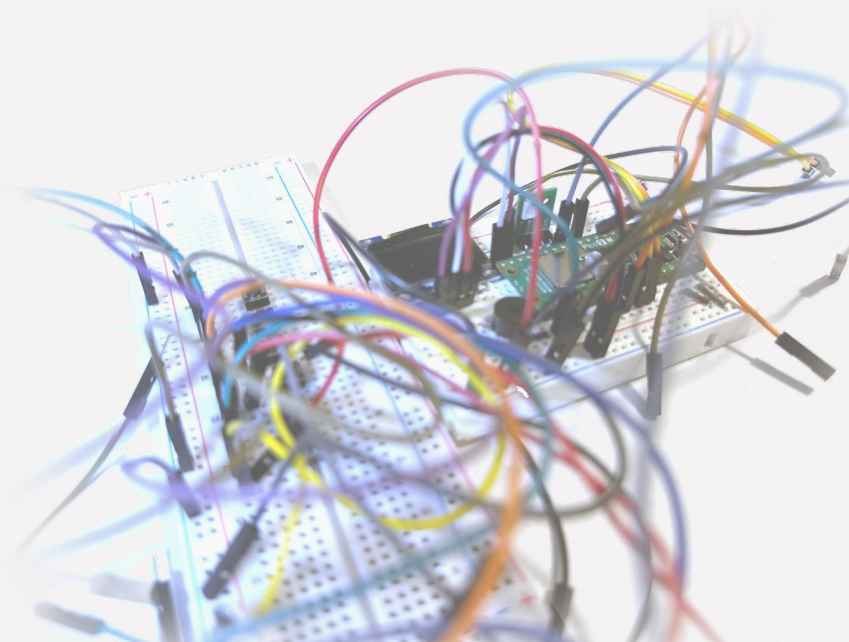
inventor pi

Introduction to Electronics:
Class Experiments Kit

Utshaho Gupta



With the right guidance,
everyone can learn
electronics, AI, and
computer vision through
projects that connect to
real-world impact.



Install Thonny app

Th

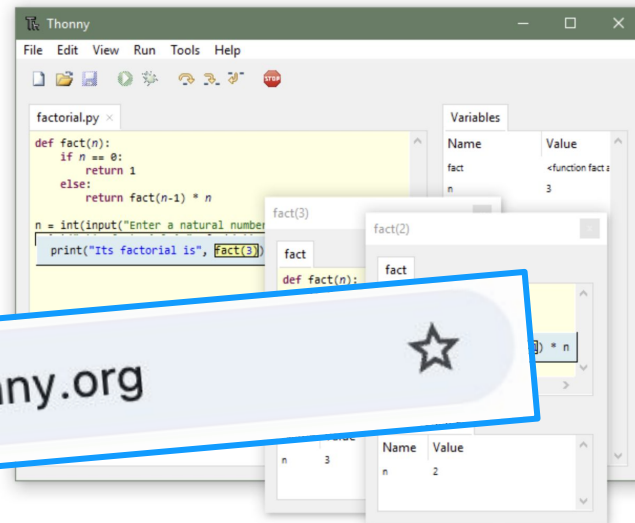
- Install Thonny app from thonny.org
- Open Thonny
 - Go to Tools > Options > Interpreter
 - Select MicroPython (Raspberry Pi Pico)
 - Click OK

Name

Thonny
Python IDE for beginners



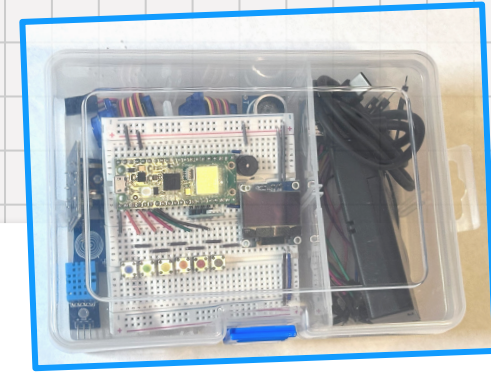
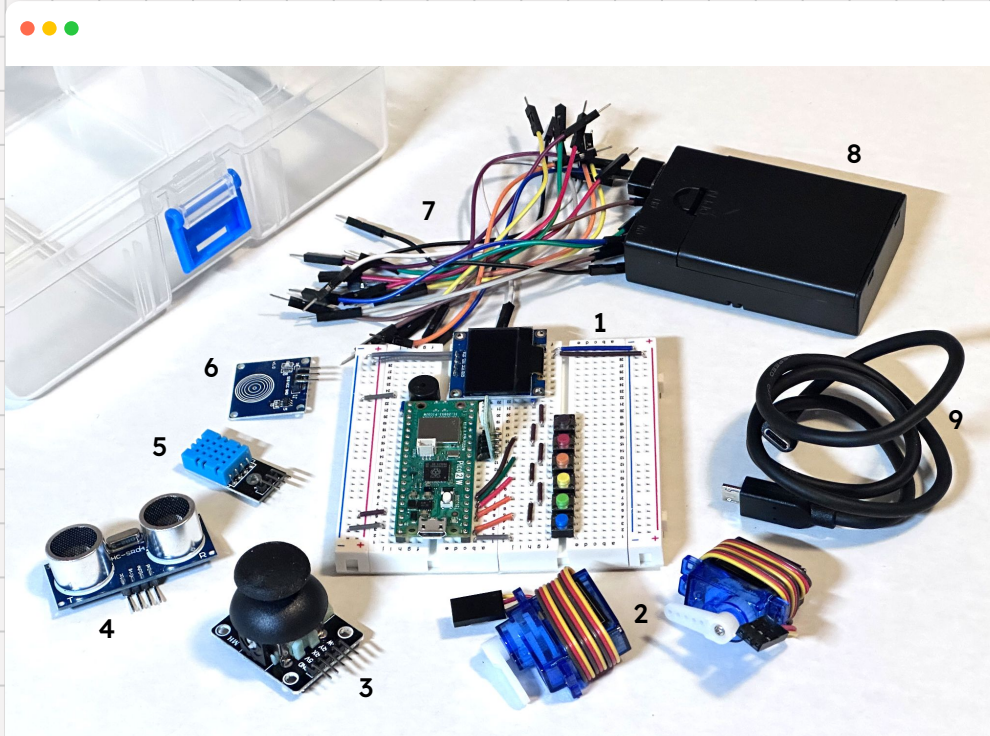
Download version **5.0.0** for
[Windows](#) • [Mac](#) • [Linux](#)



thonny.org

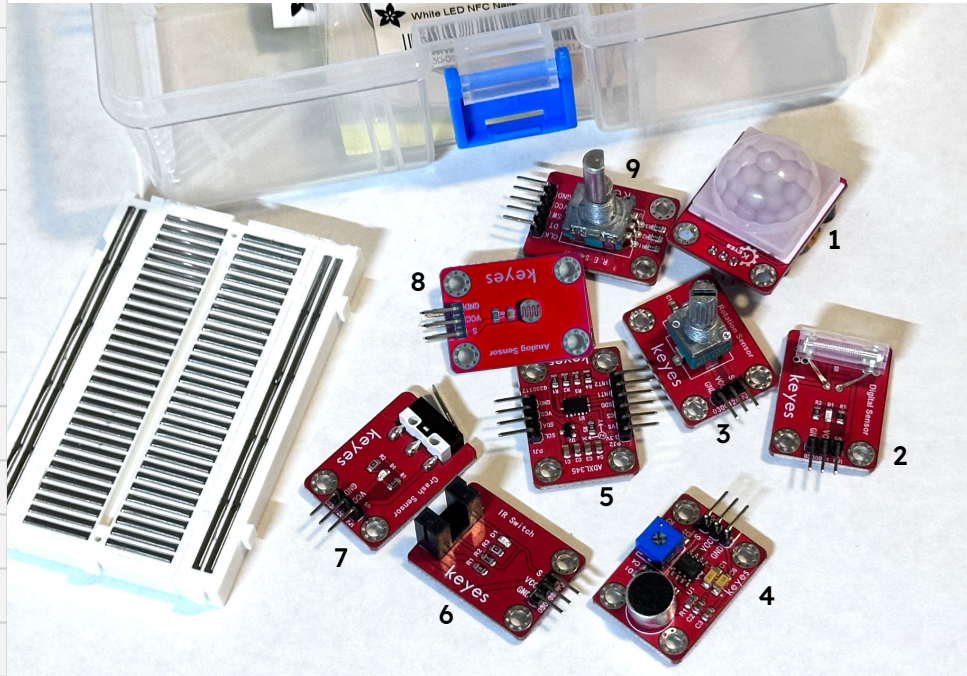


Kit Contents



1. Breadboard with OLED, LED, buzzer, Raspberry Pi Pico 2 W, 6 buttons
2. 2 Micro servo motors
3. Joystick module
4. Distance sensor
5. Humidity + temperature sensor
6. Touch sensor
7. 20 jumper wires
8. Battery pack
9. Micro USB cable

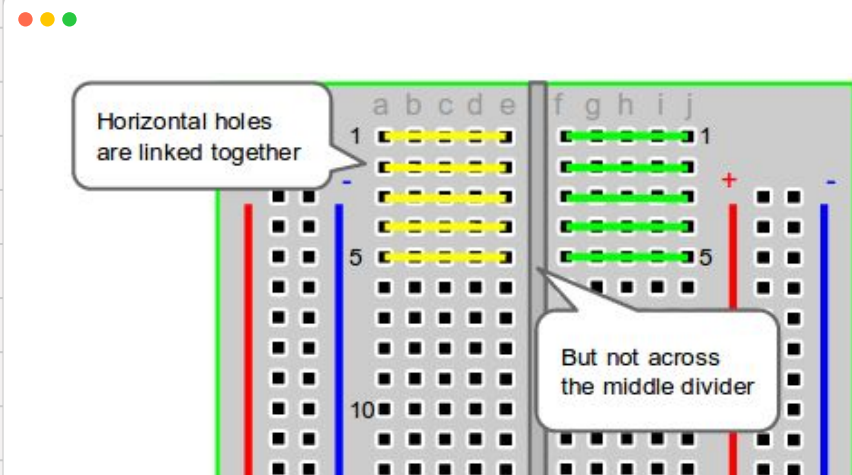
Additional Sensors



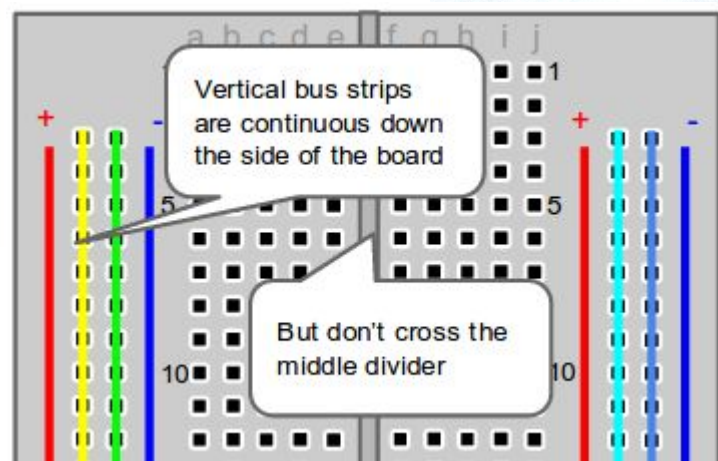
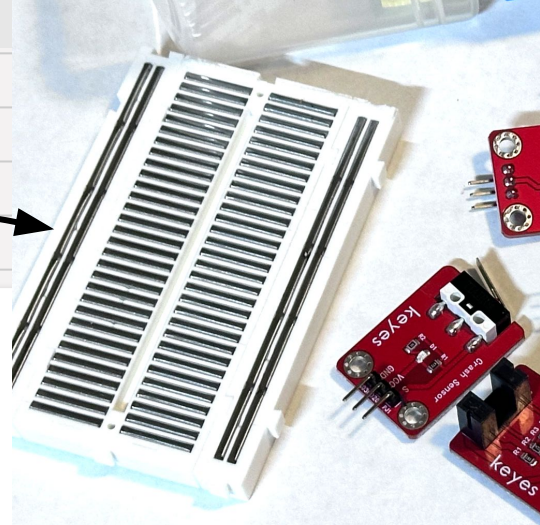
Additional Sensors for Sharing

1. Motion sensor
2. Knock sensor
3. Potentiometer
4. Microphone sensor
5. Accelerometer
6. Photo interrupter
7. Crash sensor
8. Photoresistor
9. Rotation Encoder

Breadboard Wiring

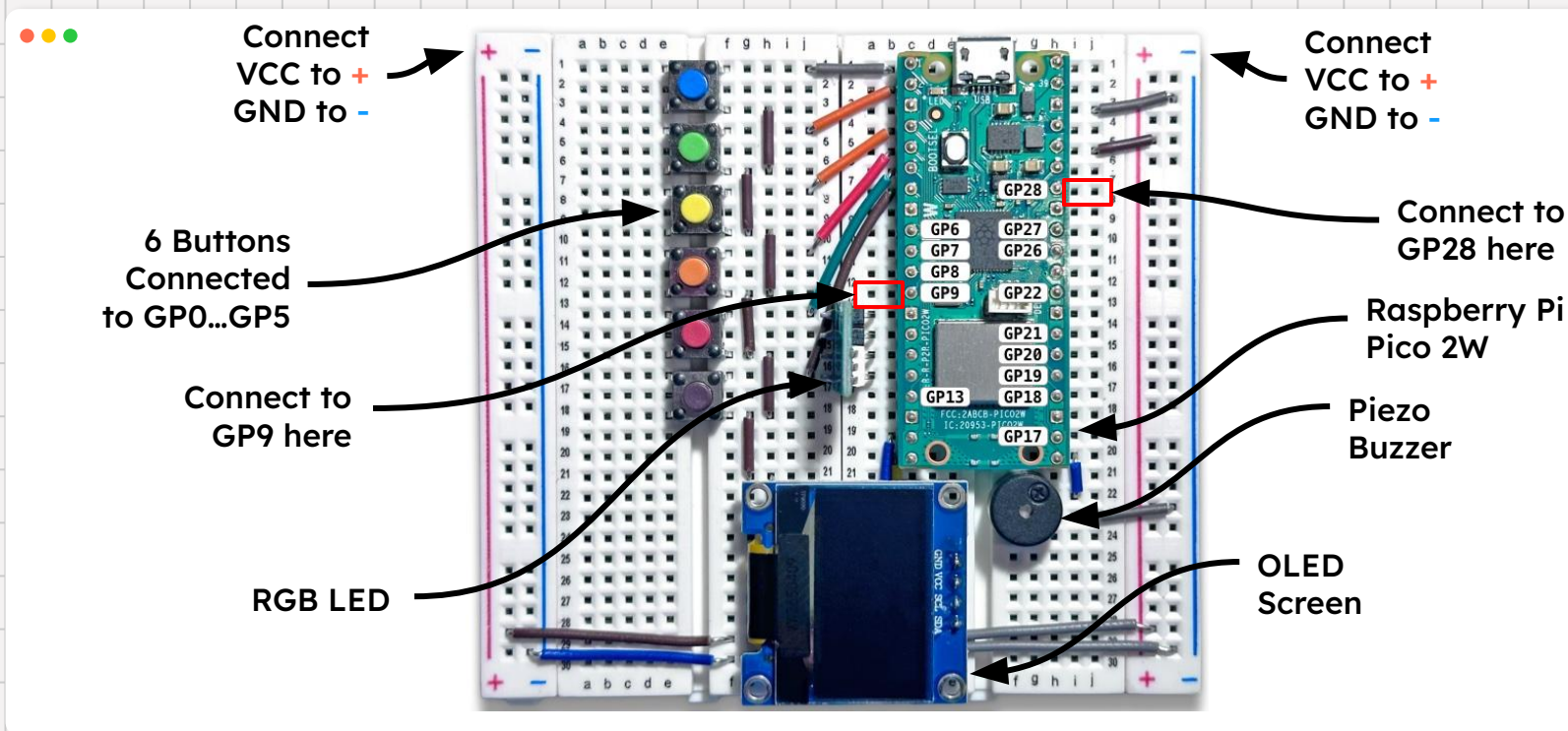


You can see the connections behind a breadboard

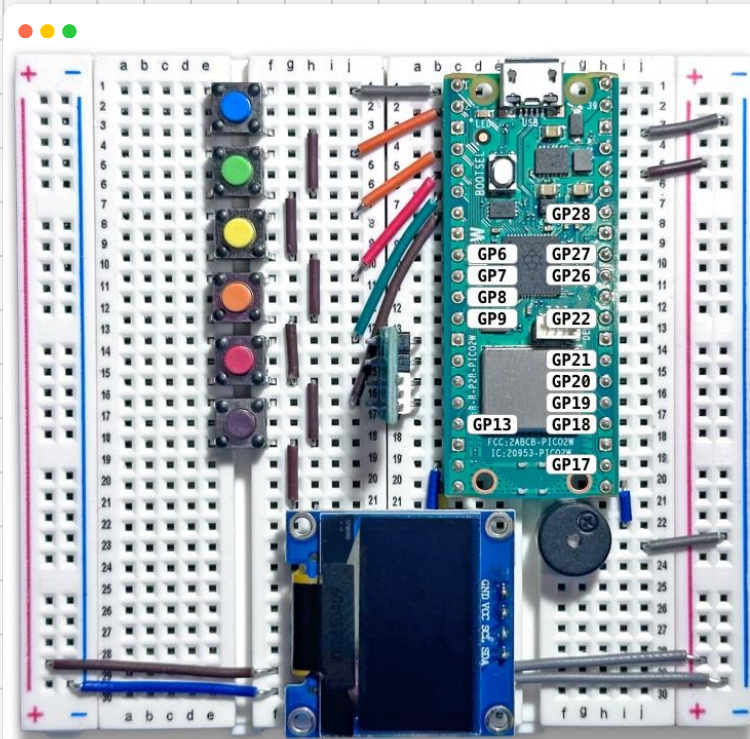


Breadboard is a solderless way to connect electronic components for prototyping

Breadboard Wiring



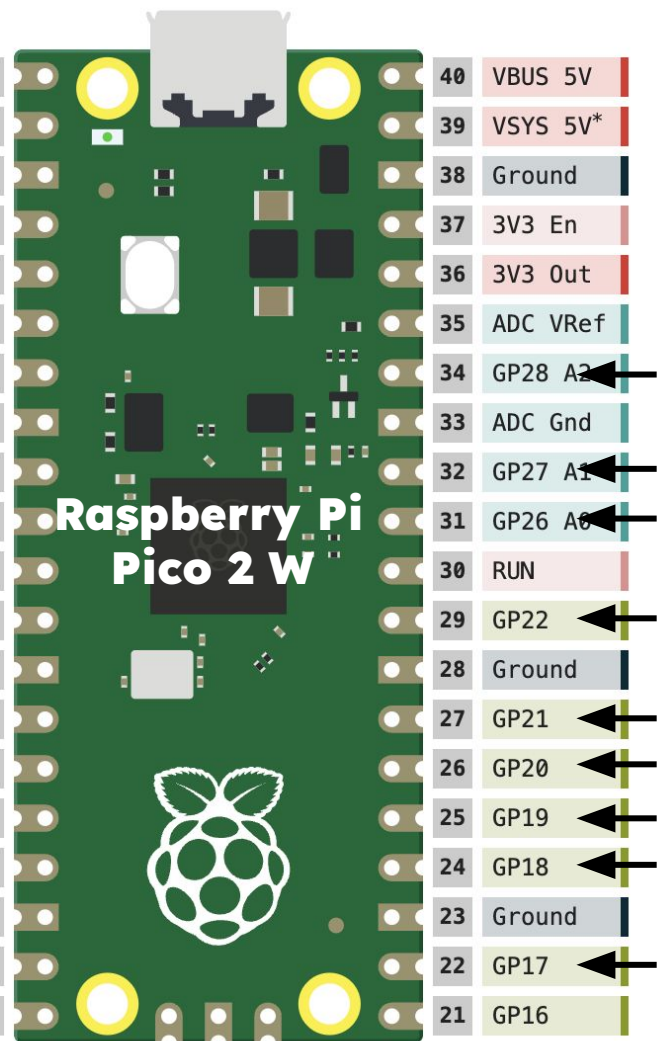
GPIO Pinout



We'll only
use the
pins with
an arrow



GP0	1
GP1	2
Ground	3
GP2	4
GP3	5
GP4	6
GP5	7
Ground	8
GP6	9
GP7	10
GP8	11
GP9	12
Ground	13
GP10	14
GP11	15
GP12	16
GP13	17
Ground	18
GP14	19
GP15	20



Important Do's and Don'ts

Do's

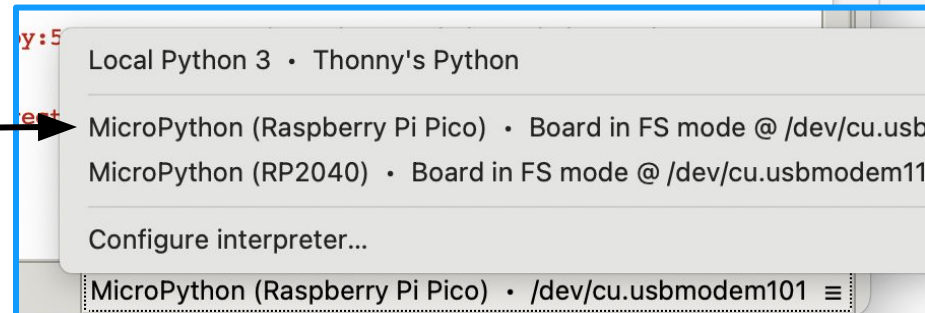
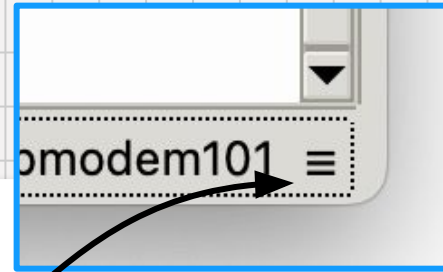
- 
1. Disconnect Pico before changing connectors
 2. Verify all connections, especially VCC and GND
 3. Establish team roles like coding, wiring, verifying circuit, capturing notes and recording
 4. Read experiment description and code

Don'ts

- 
1. Don't power on before verifying all connections. Incorrect VCC and GND can destroy sensitive electronics
 2. Don't rely on numbers printed on the breadboard. Count point distance from other connectors on the board to verify
 3. Don't leave before putting everything back in the kit box

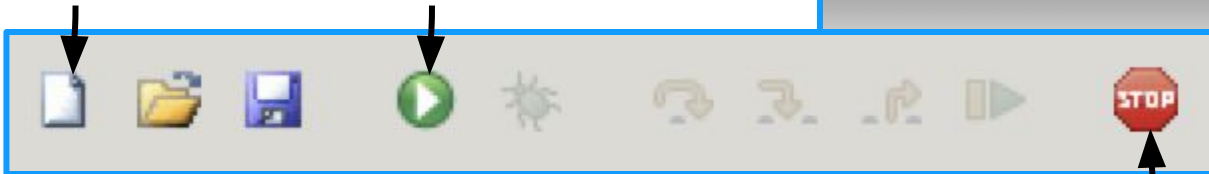
Connect Pico

1. Launch Thonny on computer
2. Connect Pico to computer via USB
3. Click  icon on bottom right
4. Select Raspberry Pi Pico



New File

Run on Pico



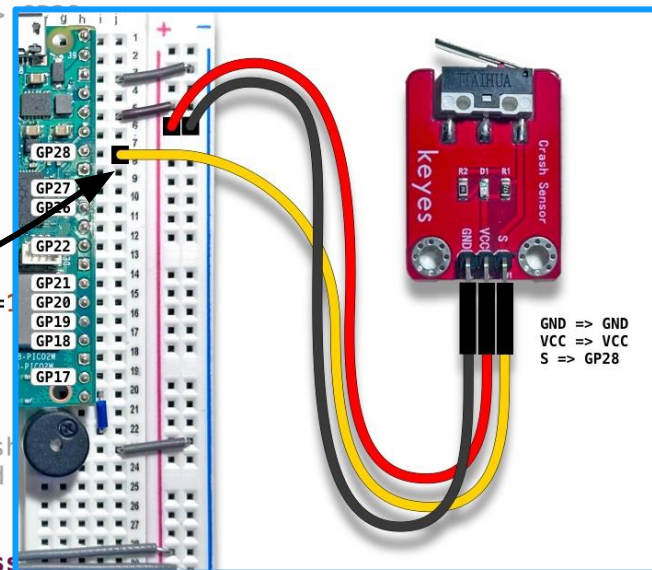
Stop Running

Code

Reading example code

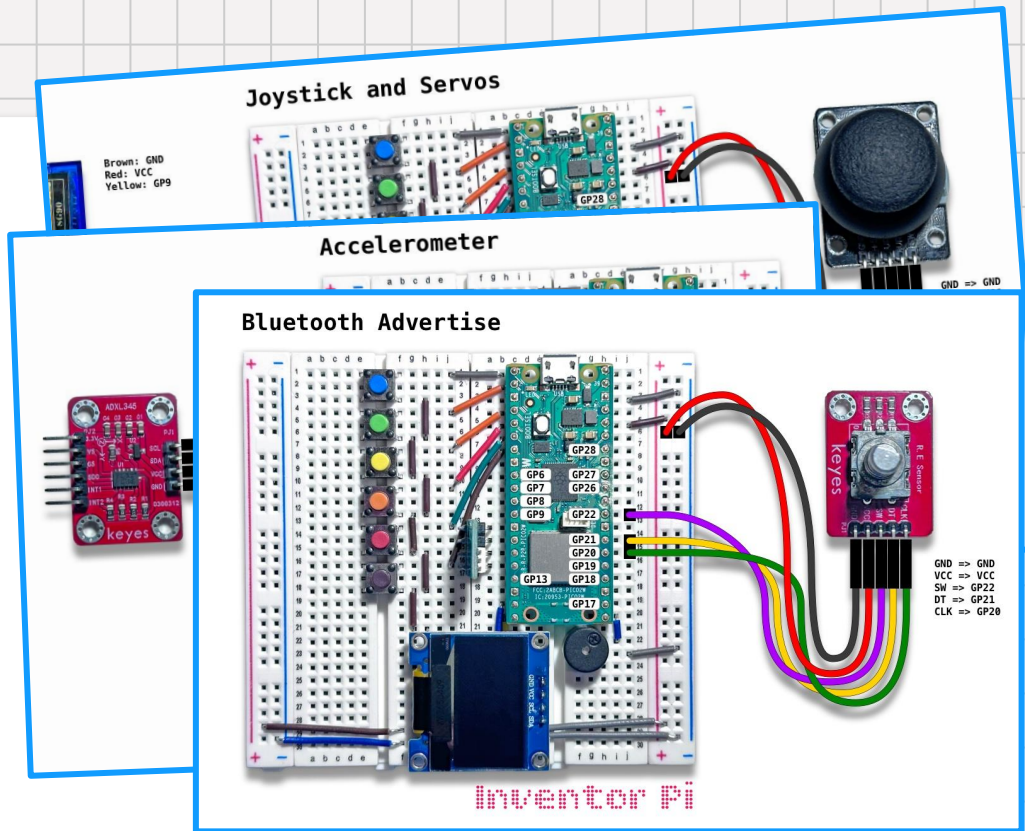
1. First read description
2. Connect any components
3. Read the main loop
4. Read other comments
5. Read rest of the code

```
crash_sensor.py
1  print("\nCrash Sensor")
2
3  # DESCRIPTION
4
5  # A crash sensor uses a limit switch. This can be used with a hard stop to detect
6  # when a mechanism, such as a robotic arm, reaches the limit of its motion and
7  # trigger a limit switch. It is also used in oven and refrigerator doors to turn
8  # on the internal light when the door is opened.
9
10 # CONNECTIONS REQUIRED
11
12 # Connect Crash Sensor (KY-021) module
13 # Pins: GND => GND, VCC => VCC, S =>
14
15 from machine import Pin
16 from oled import OLED
17 from buzzer import Buzzer
18 from rgbled import RGBLED
19
20 sensor = Pin(28, Pin.IN, Pin.PULL_UP)
21 oled = OLED(scl_pin=15, sda_pin=14)
22 buzzer = Buzzer(pwm_pin=16)
23 rgbled = RGBLED(red_pin=12, green_pin=13)
24
25 try:
26     while True:
27
28         # Sensor value is 1 => No crash
29         # Loop while no crash detected
30         oled.reset()
31         rgbled.set_color(g=255)
32         while sensor.value() == 1: pass
```

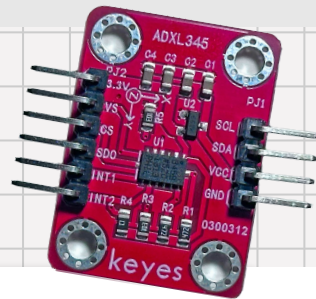


- <https://github.com/goatoush/inventor-pi>

Each team picks 1 experiment at a time. After a successful demo to the class, you can pick another experiment.



Example: Accelerometer



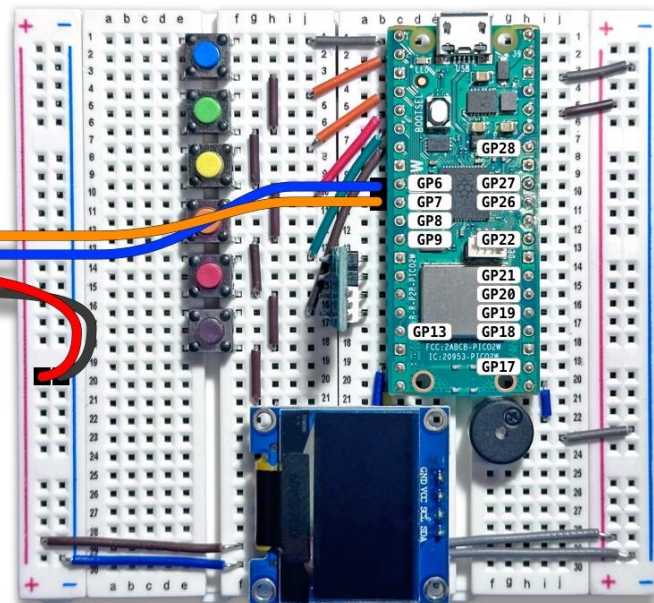
Accelerometer

An accelerometer measures acceleration around the x, y and z axis. In robotics, it is essential in determining the robot position and movement.

We will draw a 3D shape, and control its rotation using the accelerometer motion.

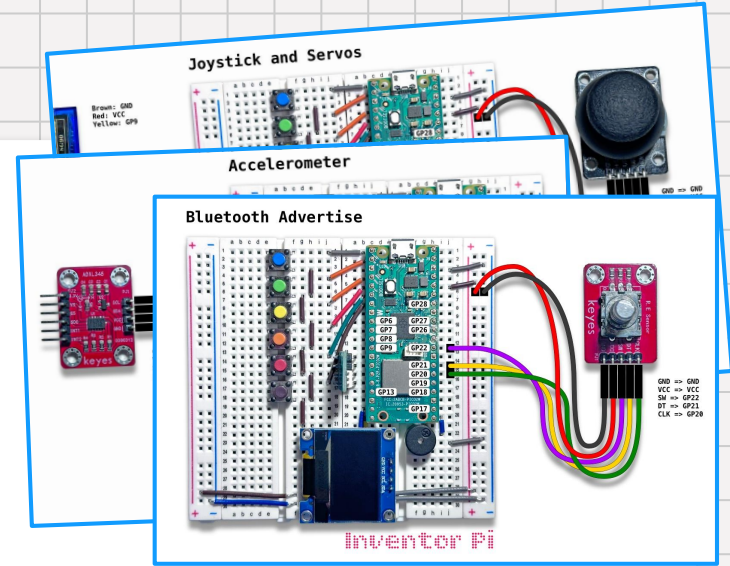


GND ⇒ GND
VCC ⇒ VCC
SDA ⇒ GP6
SCL ⇒ GP7



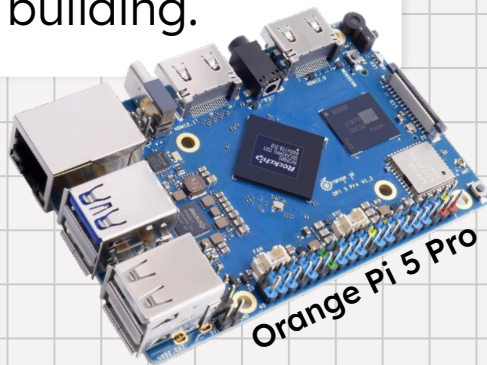
Experiment List

1. Blink
2. Humidity Temp Sensor
3. Melody
4. Touch Game
5. Distance Sensor
6. Piano
7. Joystick and Servos
8. Microphone Sensor and Melody
9. Accelerometer
10. Crash Sensor
11. Knock Sensor
12. Motion Sensor
13. Photo Interrupter
14. Photoresistor
15. Potentiometer and Servo
16. Bluetooth Advertise and Bluetooth Scan
17. Mix and match inputs and outputs



Continue Learning

We focused on learning the building blocks. You saw how easy it is to start building.

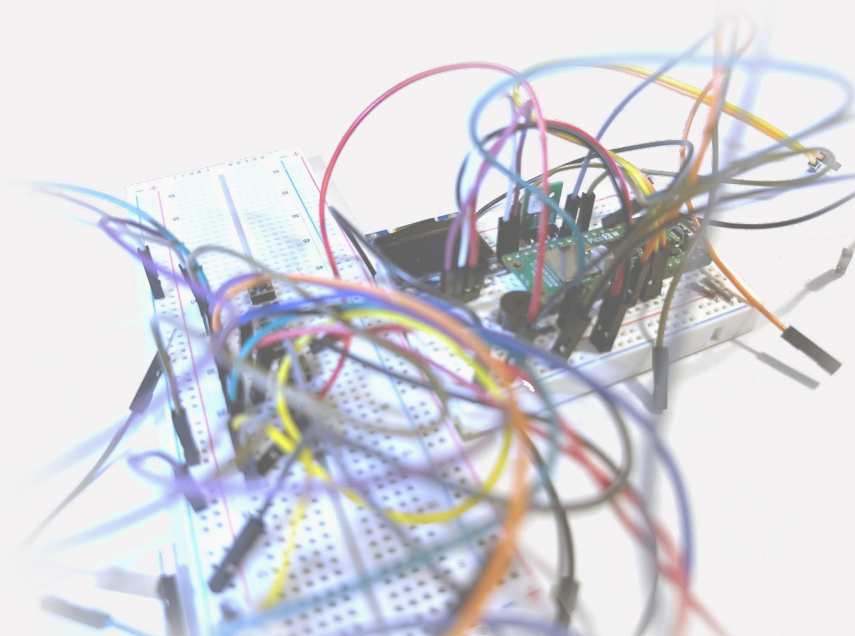


Now it's your turn to prototype your ideas. Identify real world problems, and continue learning new skills, experimenting and **inventing**.

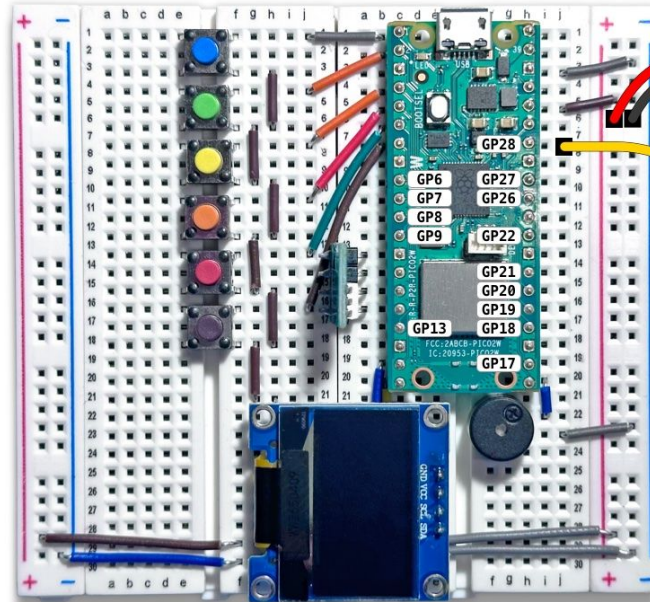


Email with your progress and any questions at ugupta@aischool.org

Experiment Circuit Diagrams for Reference



Humidity Temp Sensor

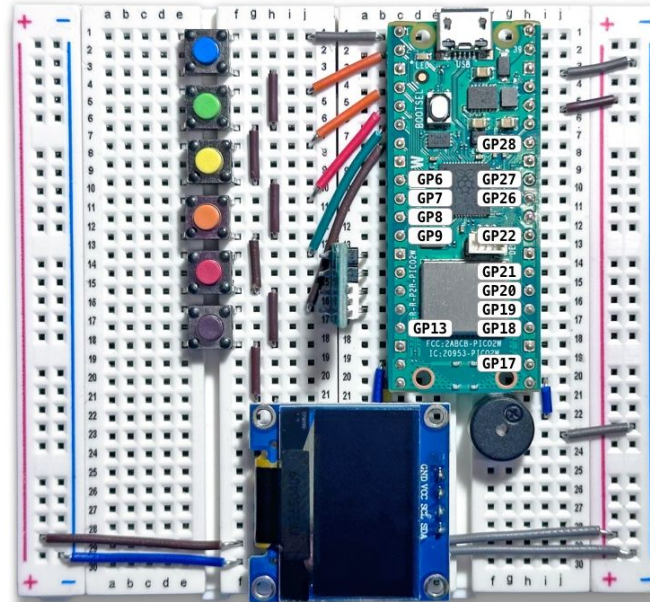


+ => VCC
OUT => GP28
- => GND

CAUTION:
Notice the
VCC and GND pins
are in different
order here.
Incorrect wiring
will damage
this module and
make it unusable.

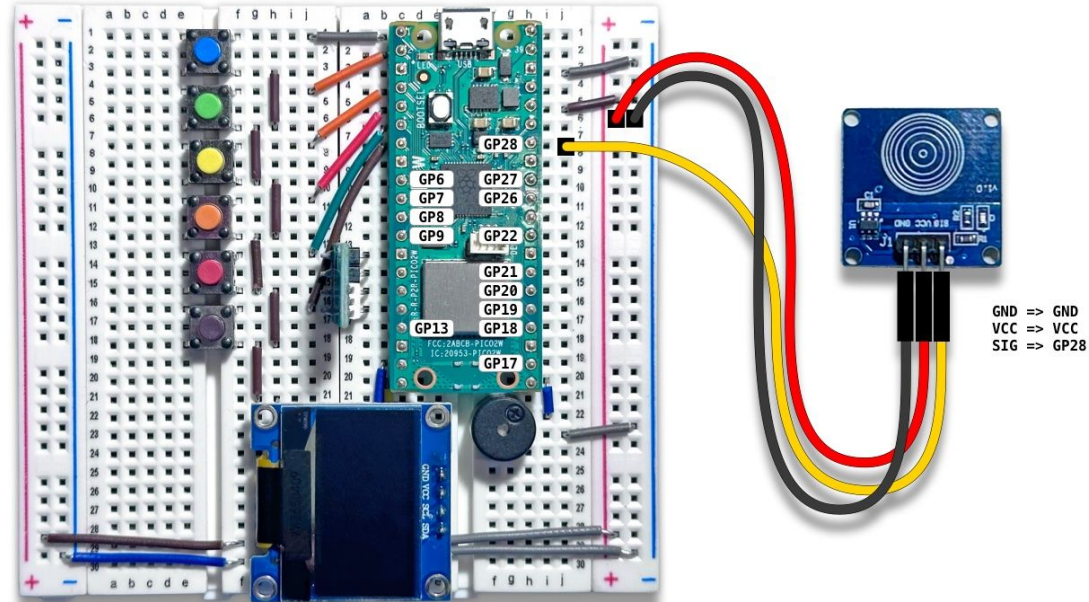
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Melody



Inventor Pi

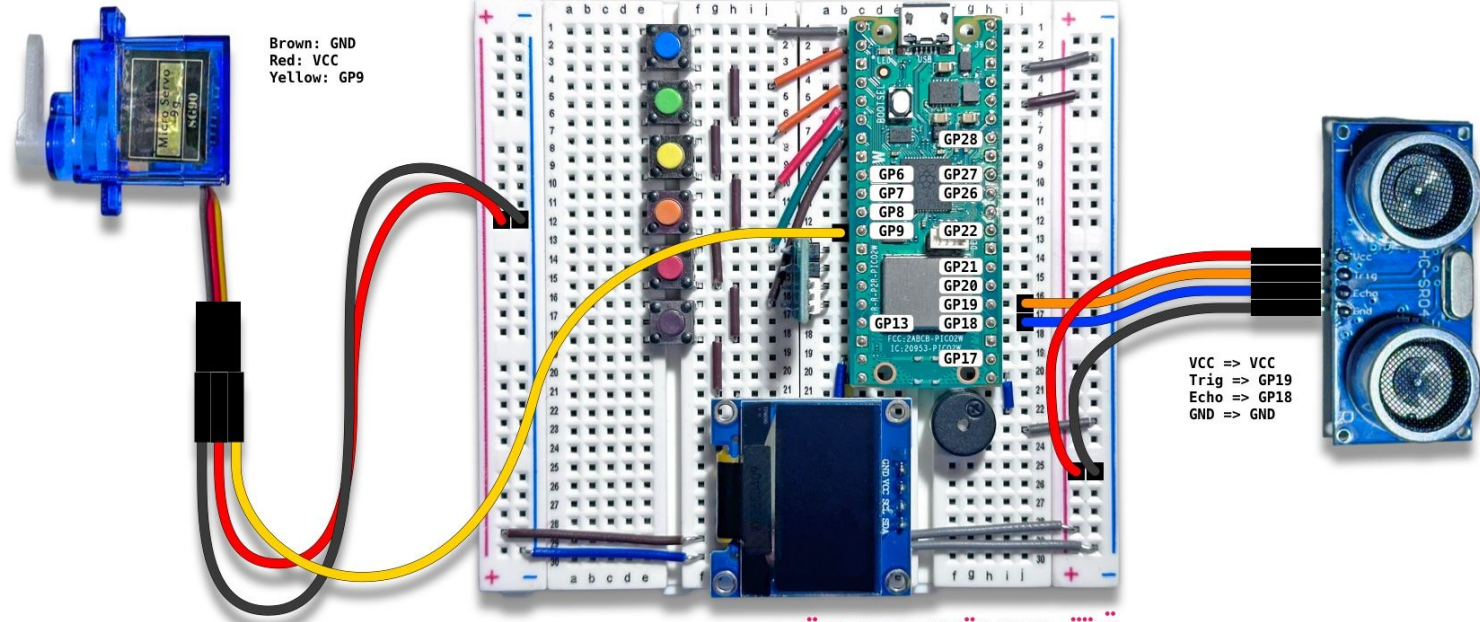
Touch Game



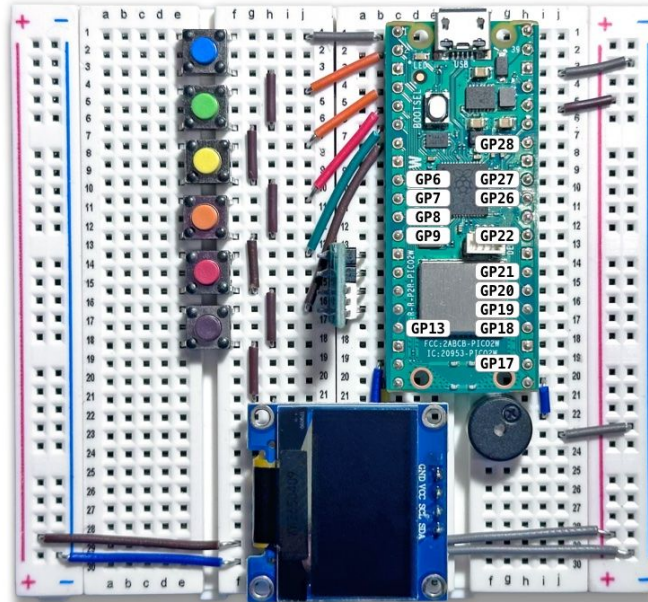
GND ⇒ GND
VCC ⇒ VCC
SIG ⇒ GP28

Inventor Pi

Distance Sensor

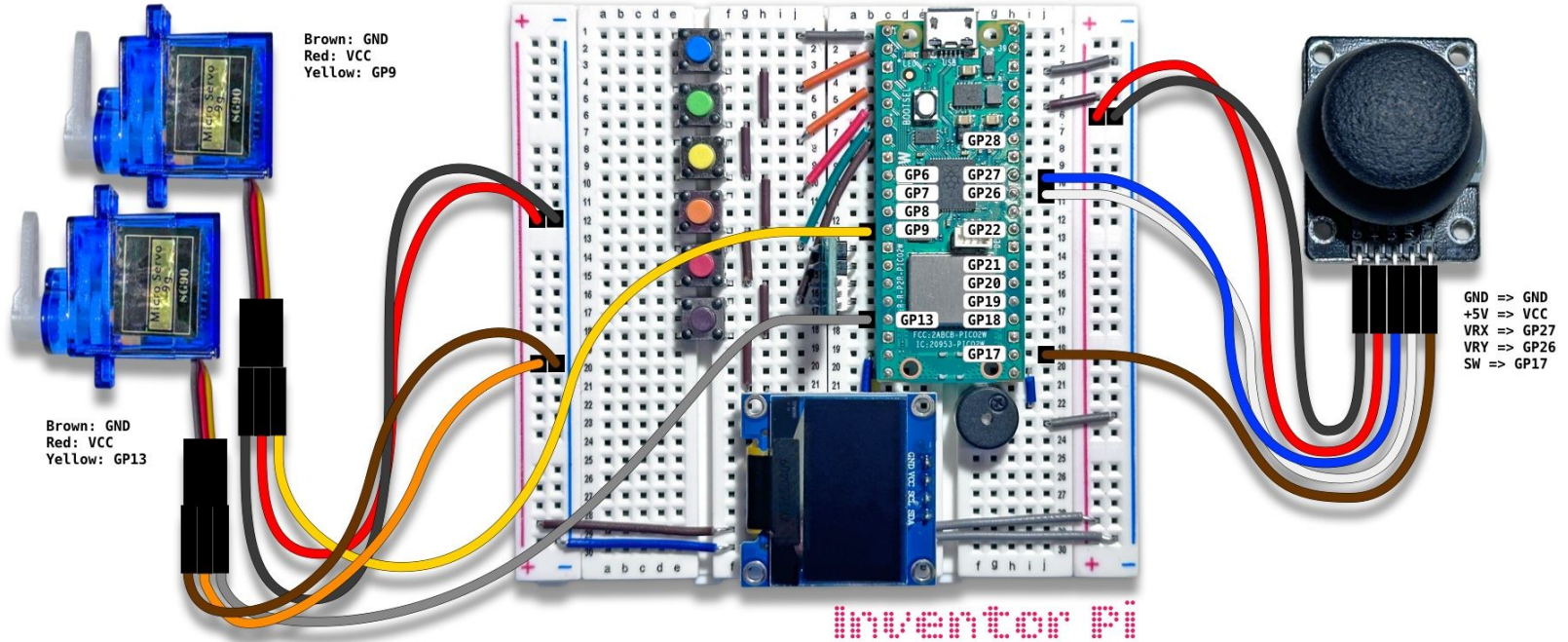


Piano

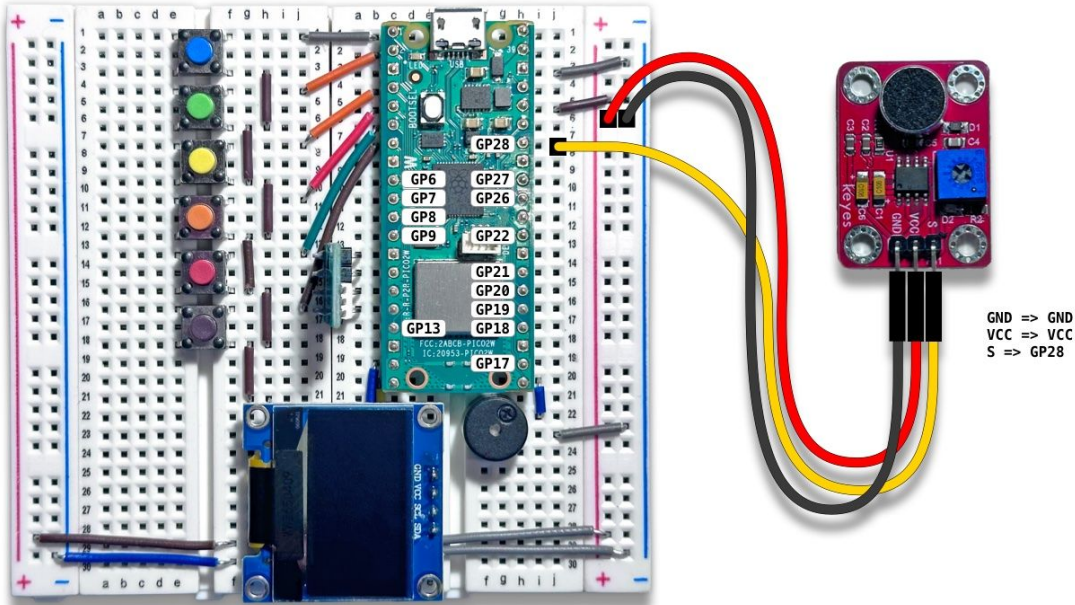


Inventor Pi

Joystick and Servos

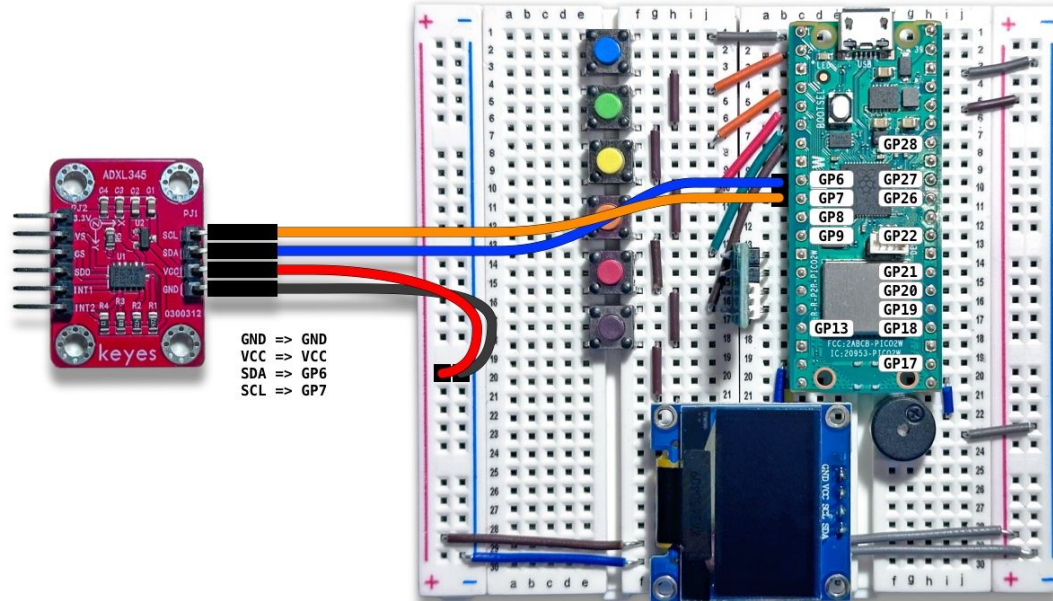


Microphone Sensor



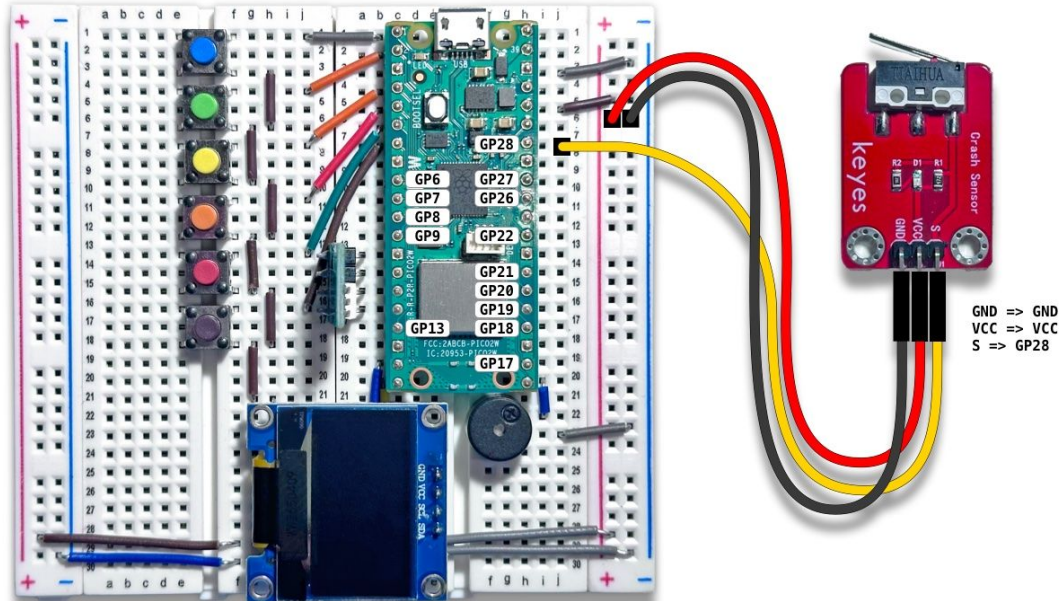
GND => GND
VCC => VCC
S => GP28

Accelerometer



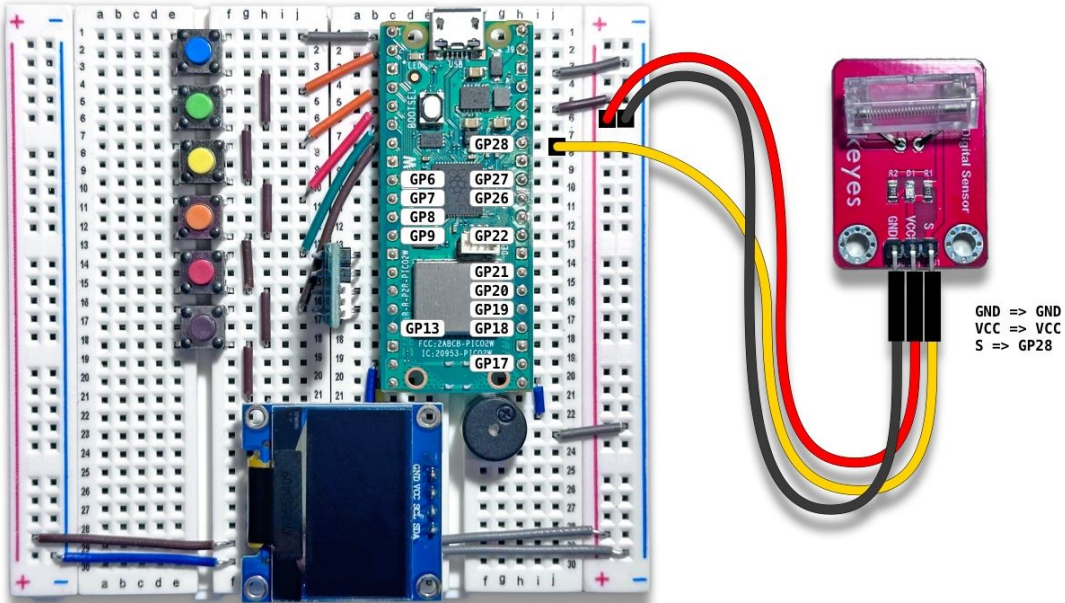
Inventor Pi

Crash Sensor



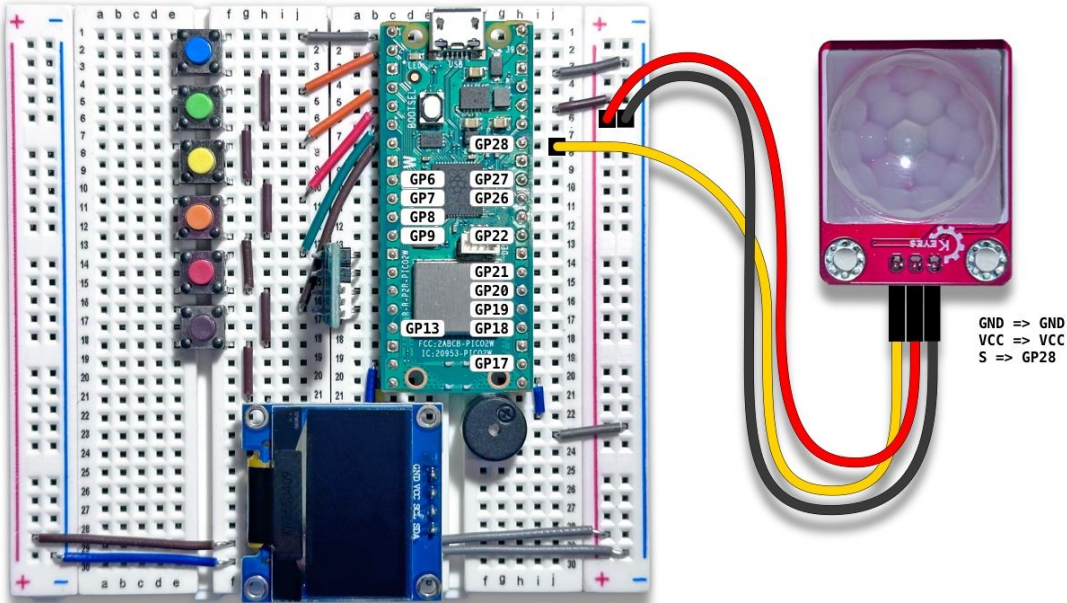
Investment

Knock Sensor



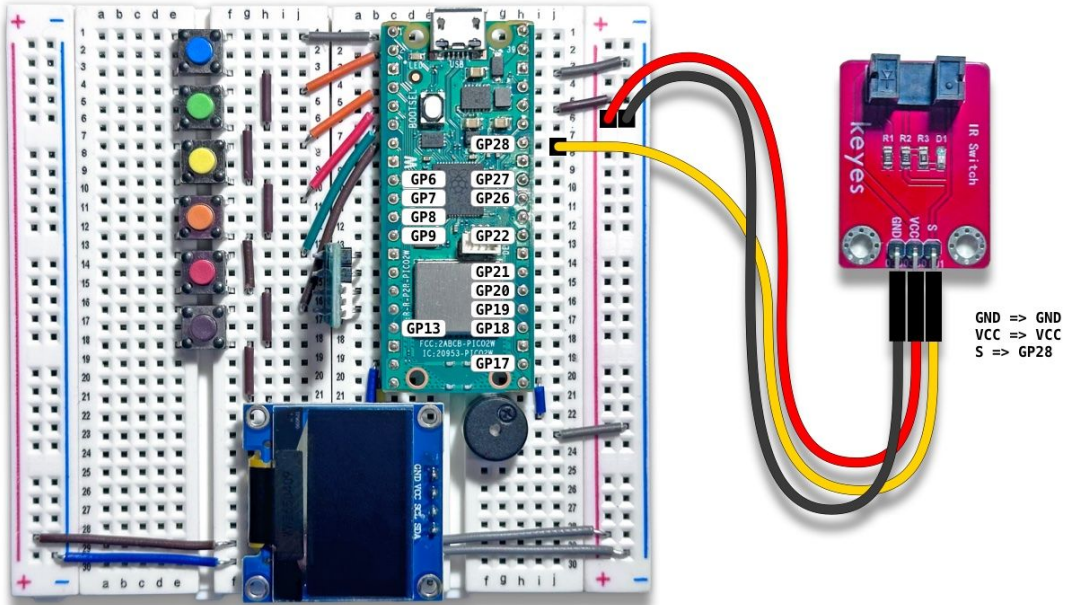
Inventor Pi

Motion Sensor



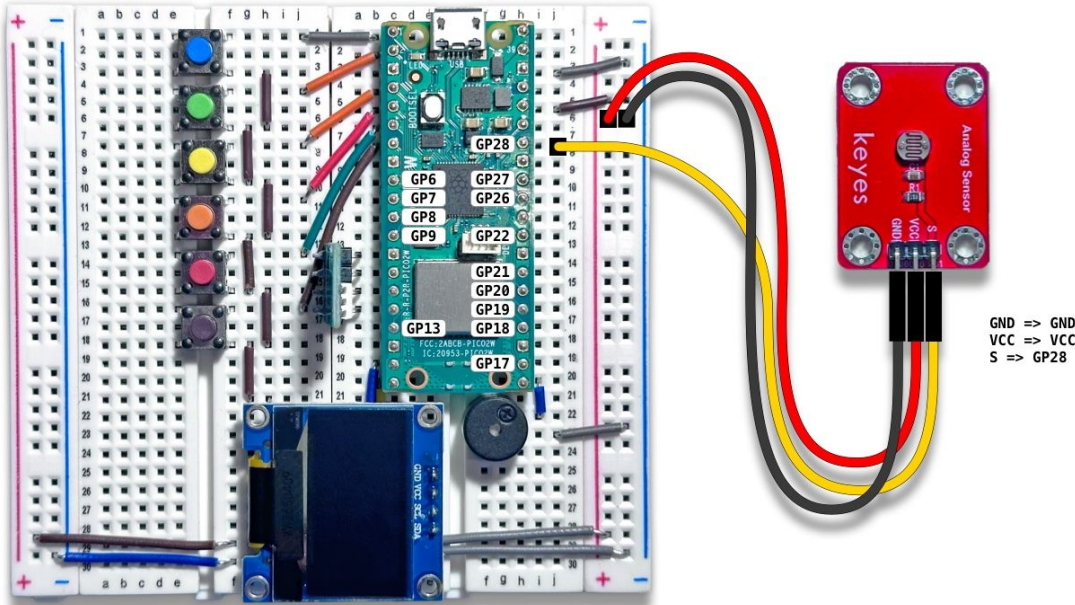
Inventor Pi

Photo Interrupter



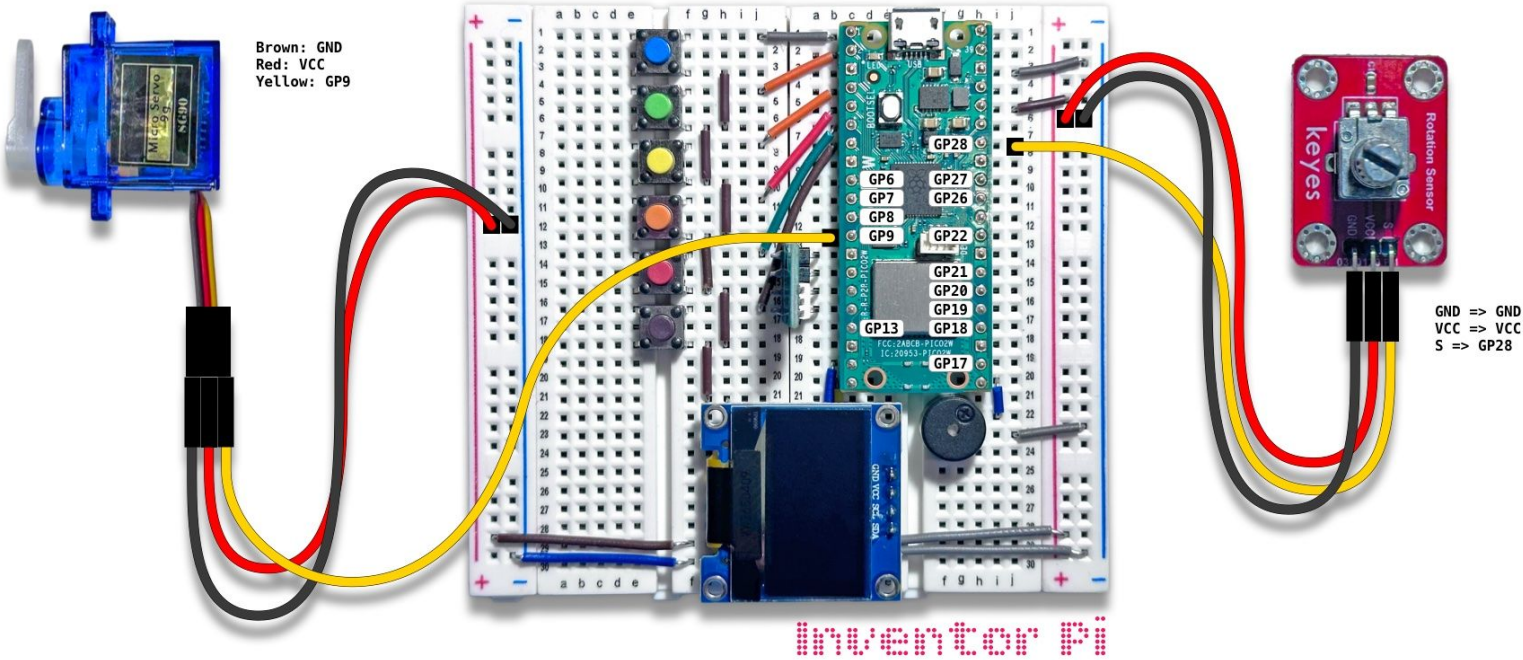
Inventor Pi

Photoresistor

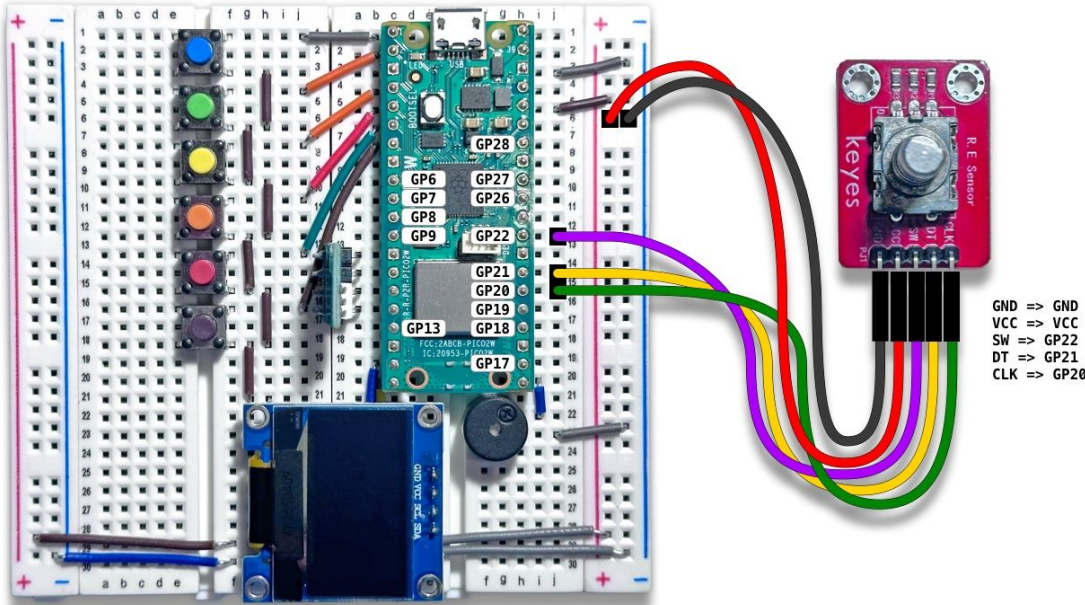


Inventor Pi

Potentiometer and Servo



Bluetooth Advertise



Inventor Pi